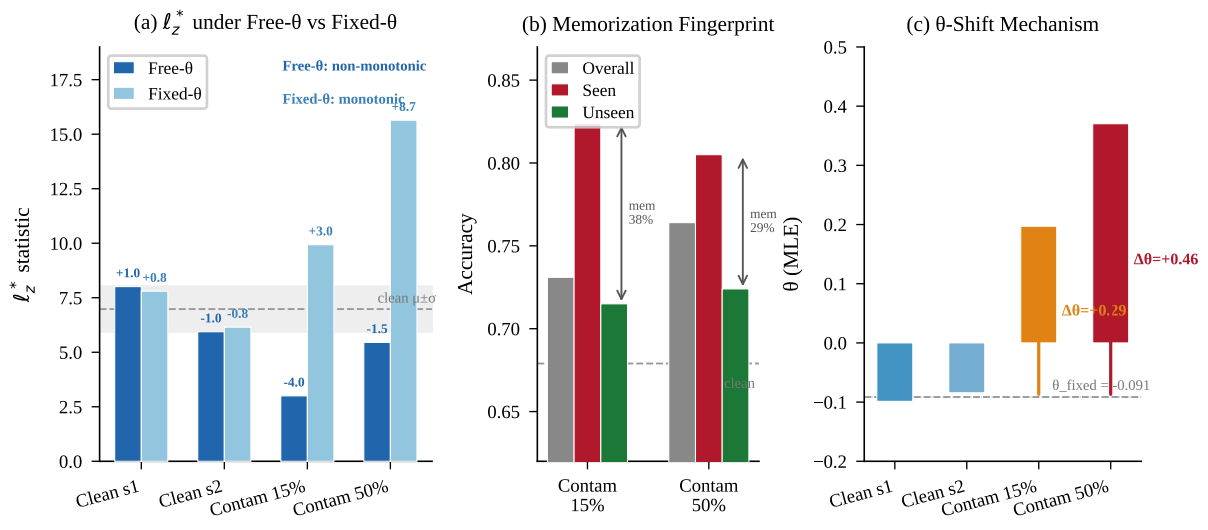
(a) ℓ_z^* under Free-θ vs Fixed-θ
Free-θ
Fixed-θ
Free-θ: non-monotonic
Fixed-θ: monotonic
ℓ_z^* statistic
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
+1.0
+0.8
-1.0
-0.8
-4.0
+3.0
-1.5
+8.7
clean μ±σ
Clean s1
Clean s2
Contam 15%
Contam 50%
(b) Memorization Fingerprint
Overall
Seen
Unseen
Accuracy
0.85
0.80
0.75
0.70
0.65
mem
38%
mem
29%
clean
Contam
15%
Contam
50%
(c) θ-Shift Mechanism
θ (MLE)
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
Δθ=+0.46
Δθ=+0.29
θ_fixed =−0.091
Clean s1
Clean s2
Contam 15%
Contam 50%